Multi-DNN scheduler — the 14 scheduling algorithms

Every policy is implemented in RTL and picked by the 5-bit scheduler_select of unified_scheduler_wrapper.sv. Area and Fmax are measured — Vivado 2025.2 out-of-context synthesis, xc7a100tcsg324-1.

Basic — 7 policies

task_scheduler.sv · select 0–6 · one FIFO task queue, selection applied at dispatch

FIFO — sel 0
first task in the queue

non-preemptive · 1,501 LUT · 106.6 MHz

LIFO — sel 1
most recently enqueued task

non-preemptive · 1,516 LUT · 111.1 MHz

SJF — sel 2
min declared burst_time

non-preemptive · 1,999 LUT · 20.3 MHz

RR — sel 3
round-robin pointer,
wraps past the end

preemptive selector · 1,594 LUT · 93.4 MHz

PRI — sel 4
max declared priority

non-preemptive · 1,663 LUT · 41.7 MHz

EDF — sel 5
min declared deadline

non-preemptive · 2,374 LUT · 19.0 MHz

LRU — sel 6
min last-access time

preemptive selector · 3,035 LUT · 19.2 MHz

Advanced — 4 policies

advanced_task_scheduler.sv · select 7–10 · 4 queue levels

SRTF — sel 7
min remaining time

preemptive selector · 1,854 LUT · 21.1 MHz

HRRN — sel 8
max (wait + burst) / burst,
by integer cross-multiply

2,941 LUT · 6.5 MHz

MLQ — sel 9
scans levels high→low; levels
never change ⇒ acts as FIFO

755 LUT · 112.4 MHz

MLFQ — sel 10
scans levels low→high; demote
on dispatch, promote at wait>100

preemptive selector · 2,295 LUT · 90.9 MHz

DNN-aware — 3 policies

own modules · select 11–13 · scheduling table + MT/CT split

AI-MT — sel 11
layer-granularity memory/compute overlap;
walks the scheduling table gated by the
MT/CT balance condition

Baek et al., ISCA 2020

3,417 LUT · 86.7 MHz

BATCH-DNN — sel 12
the AI-MT memory path, plus sub-batch
splitting and merging on a per-DNN
LIFO stack when memory is tight

Ranawaka & Stenström, EuroPar 2025

4,657 LUT · 10.7 MHz

BATCH-DNN++ — sel 13
adds distance-based MT throttling (≤5
layers), bottleneck-layer expedition,
dependency-aware compute dispatch

Ranawaka & Stenström, 2025

no current synthesis result

How they are wired

`unified_scheduler_wrapper.sv` → `multi_scheduler_wrapper.sv`(Basic + Advanced — one FIFO queue) or `dnn_scheduler_wrapper.sv`(DNN-aware — scheduling-table load, separate MT/CT outputs)

The two families have incompatible interfaces: Basic and Advanced consume a FIFO task queue, DNN-aware consume a scheduling-table load and emit separate memory-task and compute-task outputs.

RR, LRU, SRTF and MLFQ re-decide every cycle in RTL, but the dispatch FSM of multi_dnn_top is single-issue run-to-completion, so preemption cannot manifest in this integration. BATCH-DNN++ has no current synthesis result — see results/sched_chooser/hw/scheduler_hw.csv.